

Rootkits & Bootkits

Terminology

Rootkit: software that maintains privileged access by **hiding** itself and its artifacts (files, processes, network) from system tools

Bootkit: a rootkit that persists before the OS loads (in firmware/boot chain)

Kernel rootkit: loads into kernel – has God-like visibility

Rootkit hiding techniques

Technique	What it does
DKOM (Direct Kernel Object Manipulation)	Unlinks a process object from the active list – 'ps' can't see it
SSDT / system-call hooks	Redirect system calls; hide files/keys/processes from callers
Filter drivers (minifilters)	Intercept file/directory requests; a file 'never existed'
Inline hooks	Patch kernel functions directly (most advanced)
Bootkit MBR/VBR hooking	Redirect boot sector flow (old), UEFI memory corruption (modern)

Symptoms (subtle)

File hashes mismatch (system files patched)

'Blue screens' after hooking

Anti-rootkit tools see anomalies (but the rootkit hides from them too)

EDR agents that detect hooking/changes uncommonly

Detection arsenal

Memory forensics: Volatility's `!ssdt`, `!modules`, `!driverscan` compare live vs image

Kernel integrity: patchguard (Windows) detects kernel hooks → system bugchecks (so rootkits avoid it, or live in kernel memory outside PatchGuard's scope)

Secure Boot + trusted boot (hardware): verify chain; doesn't read kernel fluff but boots clean

Live kernel memory diffs: contrast a known-clean baseline with the suspicious system

Bootkit modern reality

Uses boot manager/EFI deceptions (BootHole, BlackLotus)

Secure Boot bypasses being sought – that's why Microsoft now signs/attests boot components

Bootkits must evade UEFI Secure Boot; a UEFI-rooted attack is the deepest

The honest expert line

Memory-rootkit sightings are rare in the wild (huge cost, high risk of blue-screen detection) – but the demonstration lab teaches you the mechanism, which is the same mechanism EDR and anti-cheat engineers fight.

Learn the class, not just the celebrity. If you can articulate 'how would malware hide from ps aux' you already know half of endpoint defense.

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